Nicolás Kass

Ecologist and university educator · conservation, fieldwork, data and systems

Biologist and PhD candidate in Natural Sciences, with conservation as the focus of my research. I have taught at university since 2010, from first-year students to an advanced course, and handled wild animals in the field since 2008, from Talampaya to Antarctica. I also ran a retail business for four years and built its management system with AI assistants. What ties it together: I take complex things apart until other people can use them.

La Plata, Argentina · on-site in Buenos Aires or remote github.com/nicolaskass LinkedIn

Google Scholar

EXPERIENCE

University teaching assistant · Zoology

Faculty of Natural Sciences and Museum, Universidad Nacional de La Plata (UNLP)

2010 – present

- About seven years with first-year students (General Zoology) and about fifteen in an advanced course (Zoology III: Vertebrates). The same field taught from scratch and at the level of future colleagues.
- Oral and written exams, plus continuous assessment during the course, so I can change how I teach while it still matters. When a group lacks a base concept (maths, physics, chemistry, physiology), I add it briefly, at their level.
- Led groups of 10 to 15 students on field campaigns in the Iberá Wetlands, teaching sampling methods for birds, reptiles and amphibians.
- Taught Vertebrate Zoology unpaid for twelve years before the position became paid. Designed and taught outreach workshops for children for eight years.

PhD candidate · conservation research

UNLP · Somuncurá Plateau, Patagonia

2017 – present · doctoral fellowship 2018–2024

- Spatial ecology, population dynamics and conservation of a threatened Patagonian frog (*Atelognathus reverberii*). Run as a multi-year project of my own: research design, logistics for campaigns in remote areas, field problem-solving, analysis and writing.
- Capture–mark–recapture campaigns; frogs of 3.5–4 cm marked with visible implant elastomer, and stomach flushing to study diet.
- Bayesian capture–recapture analysis and regional environmental analysis in Google Earth Engine, built with AI assistants. Four manuscripts in preparation.

Field technician · Antarctic ecosystem monitoring

Instituto Antártico Argentino · CCAMLR Ecosystem Monitoring Programme (CEMP)

September 2018 – March 2019 · pre-campaign and summer campaign

- The full CEMP protocol on Adélie and Gentoo penguins: censuses, nest sampling, arrival and fledging weights, fitting and retrieving GPS loggers and video cameras, blood sampling and stomach flushing for diet, from collection to analysis.
- A protocol written to be comparable across countries and decades. In the Antarctic a sample cannot be retaken and the season does not come back: that is where protocol discipline stopped being an abstraction for me.

Field surveyor (per project)

Latitud Sur — land surveying and topography · La Plata

2023 – 2026

- Topographic control on the widening of the La Plata–Buenos Aires highway (2026): optical levelling, GNSS RTK surveying and staking out, as-built drawings in CAD, and reports.
- Benchmark transfers on a sewer project, and took part in boundary marking and parcel status surveys.

Operations manager

Digital Discos — family record store · La Plata

2022 – 2026 · in charge of the whole business from 2024

- Ran the daily operation: customer service, sales across five channels, purchasing, stock and cash closes, with one employee.
- Sold reconditioned vintage audio with one-to-one advice to customers with no technical background: what they needed, how they would use it, and nothing they didn't need.
- Specified the business's management system and directed its build with AI coding assistants, from the data model to automated marketplace publishing. It is still in production at the business, which I am no longer part of.

External ISO 9001 consultant

IMUA · La Plata

2022

- Reviewed and designed procedures in three organizations, and supported a maintenance audit and a recertification audit, both passed.

Founder

T³ — Three Tech Thinkers · La Plata

2023 – present · part-time

- My own operations and technology consulting project. Work delivered: Latitud Sur and Digital Discos.

WHAT I HAVE PUBLISHED

Public repositories documenting how my systems and analyses are built and why. No client source code: the reasoning, what I decided, what I rejected and what each decision cost. I build with AI assistants and say so; the decisions and the review are mine.

Ecological Data Analysis

github.com/nicolaskass/ecological-data-analysis

Methods behind four manuscripts · reproducibility packages private until publication

Bayesian capture–recapture cross-checked against an established reference, with prior sensitivity analysis and a simulation of the estimator's own bias. The journal uses double-anonymous review, so the data packages will be released with a DOI once each paper is accepted.

Bayesian inference · MCMC · Python · Reproducibility

Specialty Retail ERP

github.com/nicolaskass/specialty-retail-erp

The record store's management system · still in production at the business

Architecture and decision records of a system I specified and directed with AI coding assistants: modules, domain model for unique collectible items, and a consolidation refactor over live financial data.

Django · React · MongoDB · Docker · Claude Code

Geospatial Survey Toolkit

github.com/nicolaskass/geospatial-survey-toolkit

Field tooling for a land-surveying practice

Road-works quality control, survey-code validation and raster-to-CAD conversion, for surveyors with no technical setup. The tool flags anomalies and refuses to correct them: a field measurement is evidence, and the surveyor judges it.

Python · Streamlit · DXF · GNSS

Geospatial Data Processing

github.com/nicolaskass/geospatial-data-processing

Work in progress · learning project

Where I am learning Google Earth Engine, GIS, LiDAR and photogrammetry, built with AI assistants. It includes the Earth Engine work for my papers and reference notes that grow as I learn.

Google Earth Engine · GIS · LiDAR · Photogrammetry

HOW I WORK

I teach judgement, not data

Teaching, for me, is the same cycle wherever I apply it: identify the problem, design a solution, solve it, then assess and adjust. I want the people I teach to build their own judgement and tools, and to know how to find information when they don't have it, not to repeat facts and figures. What makes me proud when I assess students is seeing them reason their way to an answer.

I use AI as a tool, not as a substitute for judgement

I built my own workflow on Claude Code, with separate agents for planning, coding, reviewing and documenting. The AI writes most of the code; I make the decisions, set the criteria and audit the result. We all have the same tools; what changes the result is the person guiding them.

I write down what a decision cost

Every decision record has a section for its downsides. A document that lists only benefits is marketing, and it makes everything around it less believable.

I know where automation should stop

In the surveying tools the software flags anomalies and corrects none of them. A field measurement is evidence, and the person qualified to judge it is the one who took it.

SKILLS

Teaching	Practical sessions and field trips with groups · oral and written exams · continuous assessment · workshops for the general public · written procedures (ISO 9001) · short videos
Field & animals	Handling wild amphibians, reptiles and birds since 2008 · blood sampling and stomach flushing · VIE marking and toe-clipping · GPS loggers and video cameras on wildlife · capture–mark–recapture · 4x4 driving
Surveying & GIS	GNSS RTK surveying and staking out · optical levelling · as-built drawings in CAD · coordinate systems and projections · Google Earth Engine (learning, with AI assistance)
Data & statistics	Bayesian inference and capture–recapture models · non-parametric methods · Python and R with AI assistance · Excel and Google Sheets
Systems	Management systems (Python, Django, React) specified and built with Claude Code · Linux · Git · LaTeX · Markdown
Management	Logical framework and stakeholder analysis (Conservation Leadership Programme, 2020 and 2022; Future Conservationist Award as team co-leader) · ISO 9001 internal auditor (Bureau Veritas, 2022)

EDUCATION & LANGUAGES

PhD candidate, Natural Sciences
Universidad Nacional de La Plata · conservation of a Patagonian amphibian
In progress · aiming to submit in 2027

Specialization in University Teaching
Universidad Nacional de La Plata · all coursework passed but one course and the thesis
2016 – present

Licenciado en Biología, Zoology track
Universidad Nacional de La Plata · five-year degree with a research thesis
2016

Languages
Spanish — native · English — advanced · German — B1 (certified) · Italian — basic

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Academic record
Co-author in journals such as Oryx, Polar Biology and Journal for Nature Conservation: Google Scholar · ORCID. Full academic CV available on request.

Last updated: October 2026 · Versión en español